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Interactions between neural decision-making circuits predict long-term dietary treatment success in obesity

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A B S T R A C T

Although dietary decision-making is regulated by multiple interacting neural controllers, their impact on dietary treatment success in obesity has only been investigated individually. Here, we used fMRI to test how well interactions between the Pavlovian system (automatically triggering urges of consumption after food cue exposure) and the goal-directed system (considering long-term consequences of food decisions) predict future dietary success achieved in 39 months. Activity of the Pavlovian system was measured with a cue-reactivity task by comparing perception of food versus control pictures, activity of the goal-directed system with a food-specific delay discounting paradigm. Both tasks were applied in 30 individuals with obesity up to five times: Before a 12-week diet, immediately thereafter, and at three annual follow-up visits. Brain activity was analyzed in two steps. In the first, we searched for areas involved in Pavlovian processes and goal-directed control across the 39-month study period with voxel-wise linear mixed-effects (LME) analyses. In the second, we computed network parameters reflecting the covariation of longitudinal voxel activity (i.e. principal components) in the regions identified in the first step and used them to predict body mass changes across the 39 months with LME models. Network analyses testing the link of dietary success with activity of the individual systems as reference found a moderate negative link to Pavlovian activity primarily in left hippocampus and a moderate positive association to goal-directed activity primarily in right inferior parietal gyrus. A *cross-paradigm* network analysis that integrated activity measured in both tasks revealed a strong positive link for interactions between visual Pavlovian areas and goal-directed decision-making regions mainly located in right insular cortex. We conclude that adaptation of food cue processing resources to goal-directed control activity is an important prerequisite of sustained dietary weight loss, presumably since the latter activity can modulate Pavlovian urges triggered by frequent cue exposure in everyday life.

1. Introduction

Obesity is an epidemic that affects millions of people worldwide and that has substantial medical consequences, including cardiovascular disease, diabetes, and cancer (Ng et al., 2014; Schienkiewitz et al., 2012). Obesity involves a dysregulation of food intake (Bryant et al., 2008),

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which is controlled by an interplay of multiple neural decision-making (DM) systems (Rangel, 2013; Crombag et al., 2008; Dayan and Seymour, 2008; Lovibond, 1983; Williams and Williams, 1969; Brown and Jenkins, 1968). These neural systems include Pavlovian and goal-directed controllers.

Pavlovian controllers trigger simple, stereotypical behaviors in response to stimuli. A typical example is the salivation response of a dog when confronted with food. Pavlovian responses can also be triggered by novel stimuli if these are predictive of a reward (Rangel, 2013; Dayan and Seymour, 2008). In this framework, cues predictive of (food) reward can acquire motivational characteristics similar to those of the reinforcers and thus trigger consumption (Robbins et al., 2008). A variety of studies have suggested that this motivational mechanism (frequently referred to as 'food wanting') is mediated by activity of the brain reward system and connected areas including the striatum, orbitofrontal (OFC) and anterior cingulate cortex (ACC), hippocampus, and fusiform gyrus (Volkow et al., 2017; Rypma et al., 2015; Jiang et al., 2015; Castro and Berridge, 2014; Rangel, 2013; Berridge et al., 2009; Ito et al., 2005). Consistently, studies using cue-reactivity tasks that were developed to measure this mechanism by contrasting pictures of food and control stimuli found that persons with obesity showed stronger food-cue specific responses in these areas than healthy controls (Martin et al., 2010; Stoeckel et al., 2008; Rothemund et al., 2007). Moreover, this hyperresponsivity is associated with less favorable body weight development across a twelve-month period (Kahathuduwa et al., 2018; Murdaugh et al., 2012; Stice et al., 2010).

Pavlovian responses are triggered automatically by stimuli and thus do not take any future side effects of behavior into account. A second, goal-directed DM system is responsible for action plans that consider future consequences. When confronted with different behavioral options, the goal-directed system compares the values for the options based on a model that reflects the probability that an action will lead to a specific outcome (Rangel, 2013). For example, this system could consider that a continuous intake of high-calorie food items can lead to undesirable body weight increases and thus inhibit the impulse to eat. The goal-directed system is typically studied using delay discounting tasks, where participants are required to exert goal-directed control by delaying the receipt of a rewarding stimulus to obtain a larger reward in the future. Delay discounting has revealed a network of brain regions underlying goal-directed control, including dorsolateral prefrontal (DLPFC) and ventromedial prefrontal cortex (VMPFC), parietal and insular areas (e.g. Guo and Feng, 2015; Rangel, 2013; McClure et al., 2004). Activity in the DLPFC measured with this task is positively linked to behavioral impulse control (Weygandt et al., 2013; Hare et al., 2009) and favorable weight development across time (Weygandt et al., 2015, 2013; Kishinevsky et al., 2012; see McClelland et al., 2016 for an overview).

Together, the existing neuroimaging studies provided valuable insights into neural mechanisms of obesity and body mass development (e.g. Weygandt et al., 2015, 2013; Kishinevsky et al., 2012; Murdaugh et al., 2012; Martin et al., 2010; Stice et al., 2010; Stoeckel et al., 2008; Rothemund et al., 2007) by either analyzing Pavlovian or goal-directed DM systems. However, a substantial number of findings suggest that dietary DM is also influenced by the interplay between these systems. For example, inhibitory interactions have been found by animal studies showing that pigeons do not learn to refrain from pecking an anticipatory Pavlovian food stimulus when refraining is rewarded (Williams and Williams, 1969). Moreover, modulatory interactions were demonstrated by studies showing that anticipatory Pavlovian stimuli strengthen the instrumental responses to discriminative stimuli when co-occurring in the instrumental context (e.g. Crombag et al., 2008; Lovibond, 1983). Please see Rangel (2013) for an overview on dietary DM and Tops et al. (2017) for a more generic model of interactions between neural systems involved in self-regulation.

Inspired by these findings, we aimed to investigate the impact of interactions between Pavlovian and goal-directed DM systems on the clinically most important treatment outcome in obesity – long-term maintenance of dietary weight loss (e.g., Sjöström et al., 1998; Wadden et al., 1998; Brownell and Wadden, 1992). To achieve this goal, we utilized fMRI data measured in 30 persons with obesity at up to five time points with a cue-reactivity and a delay discounting paradigm across the full 39-month period of a clinical research project comprising a weight reduction and a subsequent weight maintenance stage. This was done in four longitudinal fMRI analyses using linear mixed-effects (LME) regression. In the first two, we searched for brain regions showing two types of obesity-related Pavlovian responses. Specifically, given that persons with obesity show increased food wanting due to a hyper-responsivity of the reward system to food cues (i.e., 'anticipatory reward surfeit'; Adise et al., 2018; Kroemer and Small, 2016; Val-Laillet et al., 2015; Berridge et al., 2009), we searched for areas responding stronger to high-caloric food than neutral control pictures in a stable fashion across all time points in the cue-reactivity task in Analysis 1. However, given that participation in a weight-reduction program could also function as Pavlovian extinction learning since food cues become less frequently coupled to reward (e.g. Peters et al., 2009), we additionally searched for regions showing a neural response decline across the study period in Analysis 2. Given that existing cue-reactivity obesity studies (e.g. Murdaugh et al., 2012; Stice et al., 2010; Martin et al., 2010; Stoeckel et al., 2008; Rothemund et al., 2007) are not conclusive with regard to the question of whether Pavlovian responses will be stable or weakening across the course of a combined weight reduction and maintenance program, we expected that striatum, OFC, ACC, hippocampus, and fusiform gyrus could be either found in Analysis 1 or 2. In Analysis 3, we aimed to identify brain regions responding stronger during difficult than easy food decisions in the delay discounting task in a stable fashion across the study period to identify brain regions involved in goal-directed DM. In line with a neuroeconomic model of goal-directed dietary DM (Rangel, 2013), we expected to find VMPFC, DLPFC, inferior parietal cortex, pre-SMA, precentral gyrus, and insular cortex in Analysis 3. Finally, in Analysis 4 we used activity in regions identified in Analyses 1 to 3 to model dietary success obtained across the 39-month period and related dietary variables. This was done based on network parameters reflecting the covariation of longitudinal voxel activity (i.e. principal components; PCS) computed separately for each of the three individual DM systems found in Analyses 1 to 3 as well as based on parameters reflecting the interaction between these systems. Due to the high importance of DLPFC for goal-directed decision-making and striatal regions for Pavlovian responses (Rangel, 2013), we hypothesized that interactions between these two regions would have a strong impact on future dietary weight loss in Analysis 4. For modelling based on individual systems, we hypothesized to identify the areas also assumed to be involved in Analyses 1–3.

2. Materials and methods

2.1. Participants

Acquisition of participants and measurements conducted in the present functional neuroimaging study (which is referred to as 'fMRI study' in the following) were partially coupled to the procedures of a study performed by the Clinic of Endocrinology, Diabetes and Metabolism at Charité – Universitätsmedizin Berlin (referred to as 'weight loss study' in the following). In the weight loss study (Mai et al., 2018; Brachs et al., 2016), individuals with obesity first participated in a twelve-week weight loss diet and were then randomly assigned to an intervention or *ad-libitum* group given they lost at least 8% body weight during the diet (see below for details on the diet and weight maintenance program). The weight loss study comprised four longitudinal measurement time points (i.e. preceding the twelve-week diet [T-3], immediately thereafter [T0], and twelve [T12] and 18 months after the end of a diet). In the fMRI study, participants were scanned at T-3, T0, T12, and additionally 24 (T24) and 36 months (T36) after the diet. Please note that this schedule defined on a monthly scale could slightly vary by some days in the fMRI study in practice due to organizational reasons. After application of inclusion and exclusion criteria of both studies (see below), a total of 30 individuals with obesity participated in the fMRI study. Fig. 1 shows the number of data sets available for each of the four fMRI analyses.

Body weight was measured directly after participants arrived at the neuroimaging center using a calibrated scale (Kern, Balingen-Frommern, Baden-Württemberg, Germany). Body height was measured using a standard stadiometer (Seca, Hamburg, Hamburg, Germany). All participants provided written informed consent prior to study participation, which was obtained in accordance with the Declaration of Helsinki. The study was approved by the research ethics committee of the Charité - Universitätsmedizin Berlin. Each patient received 10€ for participation in a clinical screening session conducted to assess the presence of inclusion and exclusion criteria and 30€ for participation in each fMRI scanning time point. Moreover, each patient that participated in the three first measurement time points (T-3, T0, T12) obtained 200€ additionally. Please note that fMRI and behavioral data of goal-directed DM measured with the delay discounting paradigm at T-3 were analyzed independently of the cue-reactivity data in Weygandt et al. (2013) and the corresponding data acquired at T0 and T12 were analyzed in Weygandt et al. (2015).

2.1.1. Inclusion and exclusion criteria in the weight reduction study

In the weight loss study, inclusion criteria were an age between 18 and 70 years, and a BMI of at least 27. Subjects that had undergone weight changes of more than 5 kg in a two-month period preceding study inclusion or participation in another diet study in this interval were excluded. After the diet, continuation of the program was only allowed for those participants included at T-3 who reduced their body weight by at least 8% compared to T-3. If this criterion was met, participants were randomized into one of two groups at T0, i.e. an intervention or an ad-libitum group (see below for details). Moreover, the presence of hypertonia, an endocrine disease, food allergies, malabsorption, recent changes in smoking behaviour, or treatments known to affect energy homeostasis led to study exclusion in the weight loss study.

2.1.2. Inclusion and exclusion criteria in the fMRI study

Inclusion in the weight loss study was the prerequisite for participation in the fMRI study. Participants were excluded when the screening with a standardized clinical interview for mental disorders indicated a borderline personality disorder or substance abuse or a delusional, anxiety, or affective disorder. Moreover, when they were ever diagnosed with dementia, Multiple Sclerosis, Parkinson's Disease, stroke, epilepsy, brain tumor. Importantly, although not explicitly defined as an exclusion criterion, screening of anatomical brain scans of all participants and timepoints by a neuroradiologist did not indicate the presence of a traumatic brain injury in any of the participants.

2.2. Body weight modification

2.2.1. Diet protocol: eight-week very low calorie diet, four-week calorie restricted standard diet

The diet conducted by the Clinic of Endocrinology, Diabetes and Metabolism at Charite – Universitätsmedizin Berlin lasted for a total of twelve weeks and comprised two stages: an eight-week very low calorie diet (VLCD) period and a subsequent calorie-restricted standard diet with a length of four weeks. During the VLCD, participants received five portions of a formula diet (Optifast 2®, Nestlé HealthCare Nutrition GmbH, Frankfurt am Main, Germany) equivalent to a daily energy intake of 800 kilocalories, the standard diet (1500 kilocalories per day) was composed in accord to guidelines of the German Association of Nutrition (55% carbohydrates, 30% fat, 15% protein; see http://www.dge.de). In line with recommendations formulated by Yumuk et al. (2015), the VLCD was applied under medical supervision. Specifically, participants made weekly hospital visits during the VLCD in which the compatibility of the diet was checked by a physician in a medical monitoring procedure and during which the participant met a nutritionist to support and protocol the weekly eating behavior. The medical monitoring procedure included a blood test, an arterial blood pressure measurement, and a patient interview. Based on the blood sample, parameters for calcium, sodium, potassium, triglycerides and creatinine were determined to guarantee a compatible course of the VLCD in terms of e.g. renal and cardiac functioning. During the standard diet, participants were provided with cooking recipes and advice regarding eating behaviours. Finally, across the full twelve-week diet period, participants met in weekly group meetings comprising cooking courses and participation in guided physical exercises (i.e., aqua fitness or therapeutic exercise training) with a

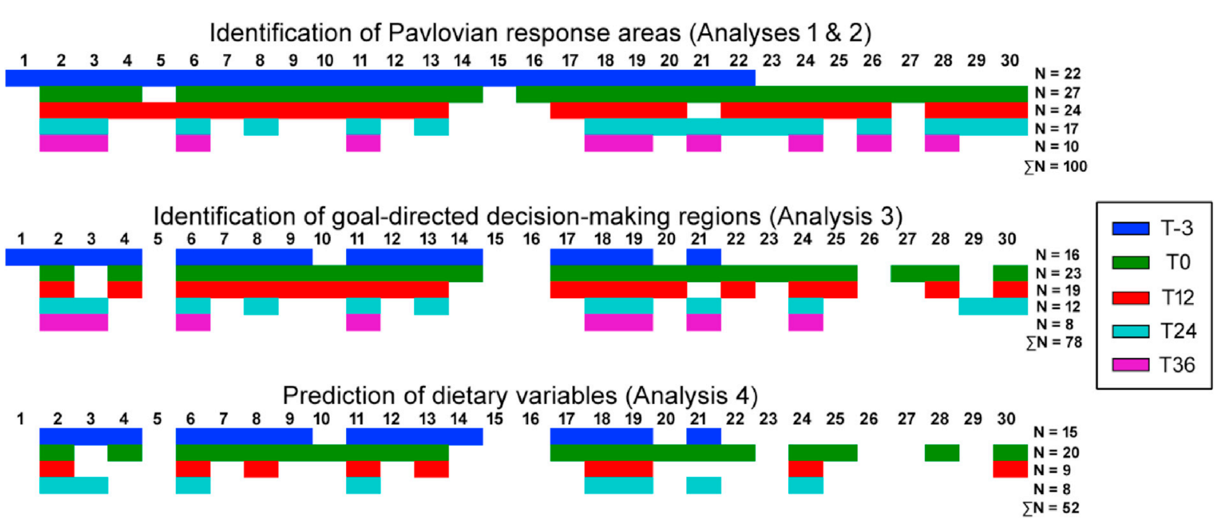

Fig. 1. Illustrates the number of data sets (i.e. fMRI, clinical, demographic, and behavioural data for individual participants and time points) available for each research aim or analysis respectively. Each colored box reflects one data set of a participant and time point. In particular, 100 data sets sampled across 30 participants and 5 time points were available for voxel-wise Analyses 1 and 2. Moreover, 78 data points were available for Analysis 3. Finally, Analysis 4 relied on 52 data sets. The lower number of data sets available for the analysis of goal-directed decision-making compared to Pavlovian responses depended on the fact that some participants showed invariant or inconsistent decision-making behaviour in the delay discounting task. When this was the case, the data of this participant and time point could not be included (see Analysis 3 for details). Moreover, the number of data sets available for prediction of dietary variables was smaller than for the analysis of goal-directed decision-making since changes in body mass for a participant and time point could only be computed if the body mass information for this and the consecutive time point was available. Thus, the number of data sets that could evaluated for each participant reduced by one (see Analysis 4 for details).

duration of 30 min. For further details, see Weygandt et al. (2013).

2.2.2. Weight maintenance program

In a weight maintenance program included in the weight loss study, participants allocated to the intervention group met in regular group meetings in the period from T0 to T12 that had a duration of 90 min. In these group meetings, subjects received nutritional counselling and participated in moderate physical exercises. The group sessions were held on a weekly basis during the first four month of the maintenance period, every two weeks during the following two month, and finally on a monthly basis during the remaining time. Nutritional counselling was provided in all group meetings and focussed on recommendations advocating the preferential intake of specific foods such as cereals, vegetables, low-fat fish and meat. Guided physical exercise identical to that performed in the twelve-week diet was performed in the first three month of the maintenance period. Although no supervision was provided during this interval, participants of the intervention group were encouraged to continue with the physical exercise at least twice a week during the rest of this period. To facilitate self-monitoring and increase motivation, the participants were also provided with pedometers. Moreover, a free three-month membership in an external gym (paid from funding of the weight loss study) was offered. Finally, members of this group also participated in psychological relaxation exercises monitored by a clinical psychologist during the first six group meetings. Contrarily, participants of the ad-libitum group only received a brief advice leaflet at T0.

2.3. Stimuli and tasks

In this study, we used two fMRI tasks: A cue-reactivity design, where participants were passively exposed to food cues to measure neural mechanisms of Pavlovian responses and a delay discounting task to study the mechanisms of goal-directed control on active food choice. Both tasks were measured between 8 and 11 a.m. to control for potential circadian variations of psychobiological factors. Participants were told to stop eating at 8 p.m. on the day before fMRI scanning to guarantee comparable satiety levels across participants.

2.3.1. Cue-reactivity task

In this task, pictures of four categories were shown. Specifically, high-calorie food pictures (having approx. 395 kcal and ≥4 kcal/g energy density on average), pictures of low-calorie foods (approx. 179 kcal and 1.5 kcal/g on average), food utensils, and neutral control stimuli (natural scenes) comprising ten pictures each were presented in a block-wise fashion. Both food picture categories contained food items that are frequently eaten at breakfast as well as at lunch. The pictures were taken from the International Affective Picture System (Lang et al., 2005) and from the internet. The four small pictures in the left of Fig. 2 depict exemplary stimuli that reflect the pictures included in the four categories in terms of energy density, content, etc. (please note that the true pictures used in the task can not be shown here for copyright reasons; Lang et al., [2005]). In each block, each picture was presented for 3000 ms followed by presentation of a fixation cross shown for 487 ms (we copied this parameter from Rothemund et al., [2007] for comparability reasons). Blocks of the three food-related categories were presented twice, followed by the presentation of a control block, which was thus shown six times. The sequence of images in a block and the sequence of food-related categories across the course of the experiment was determined in a pseudorandom fashion. The full duration of the experiment was 6 min 58 s. See Fig. 2 below for an overview.

Directly after scanning, the participants used a visual analogue scale to rate the urge the food pictures induced to consume the depicted foods and the induced appetite. Moreover, they rated their valence, arousal, and dominance using a computerized version of the Self-Assessment-Manikin (SAM) approach, which is a widely established and very reliable pictorial technique for measuring affective stimulus attributes (e.g. split-half reliability $r > 0.90$ for valence and arousal; Lang et al., 2005; Bradley and Lang, 1994). Its concurrent validity (which also depends on the properties of the reference assessment instrument) is reported to be in the range of $0.56 \geq r \geq 0.87$ (Nabizadeh-Chianeh et al., 2012).

2.3.2. Delay discounting task

In this task, participants selected several times between an immediately available small meal and one of four larger versions of the same meal available after one of ten different delay levels. To adopt to personal preferences, participants could select the meal type (tomatoes with mozzarella, eggs and tomatoes, cereals with dried fruit, pasta salad, muffins, scallop, and pizza) at their first study visit. Two fMRI delay discounting runs measured each time point were evaluated per participant. The ten delay levels were determined as the deciles of a maximally tolerated delay level assessed in a behavioral pilot plus ten percent. Each of the four delayed meal size levels was paired with each of the ten delays, resulting in a total of 40 trials. Each of the trials started with an inter-trial interval (ITI; 4–8 s) followed by the choice stage in which a participant had maximally 6 s time to select between the two options using a button box. After the choice, the feedback stage started (duration: 6 s minus choice time) in which the chosen option was highlighted by presenting a red arrowhead underneath its image. Then, the next trial started. The position of the delayed stimulus, its magnitude and delay, and the duration of the ITIs were determined in a pseudorandom fashion for each run of the task, which had a duration of 8 min each. Participants were instructed that reward would be hypothetical (cf. Weygandt et al., 2013; Doran et al., 2007). Please see Fig. 3.

2.4. Dietary variables: longitudinal body mass changes, urge to consume and valence of high-calorie foods, behavioural impulsivity

To determine a marker for longitudinal body weight changes, we computed the difference between temporally consecutive pairs of body mass index (BMI) scores (e.g. for T0 - T-3, T12 - T0, T24 - T12, and T36 - T24 if a patient participated in all five measurements) and refer to this marker as ΔBMI in the following. These difference values where then

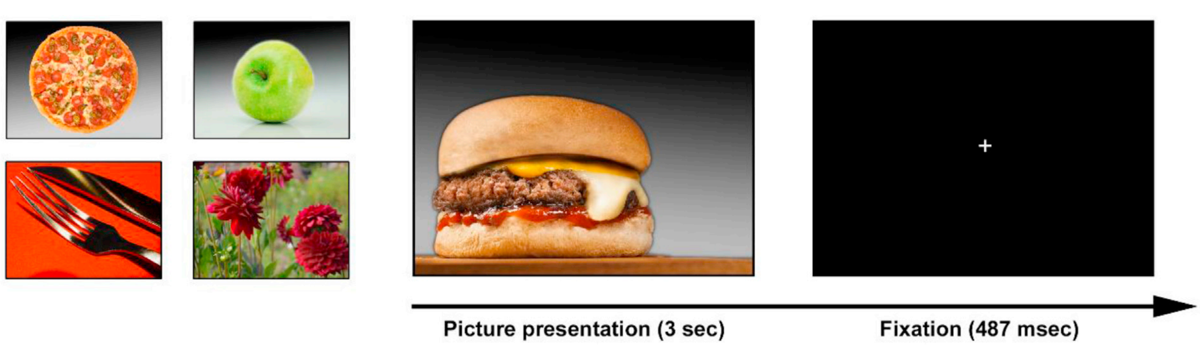

Fig. 2. Illustrates the cue-reactivity task used to measure neural substrates of Pavlovian responses.

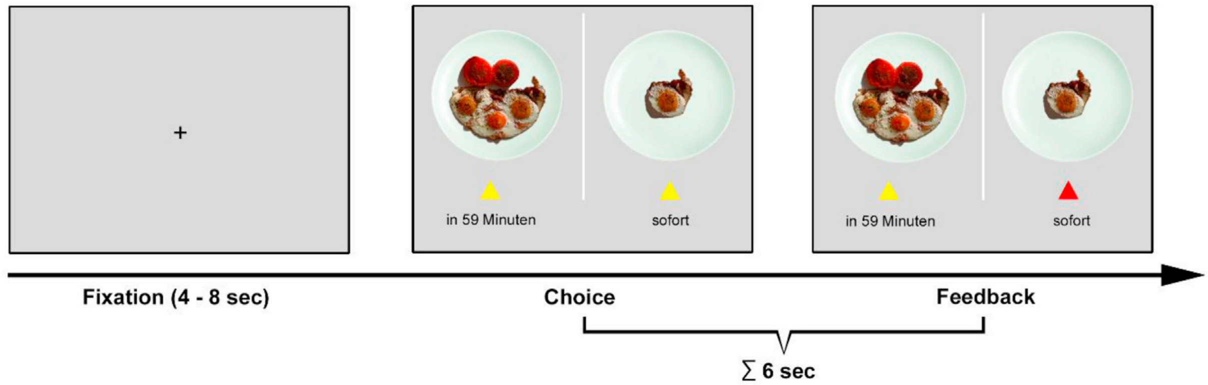

Fig. 3. Illustrates the delay discounting task used to assess neural substrates of goal-directed DM.

concatenated across all participants to form a ΔBMI criterion vector for Analysis 4 (see below) that was performed to predict changes in body mass across time. Next, we determined the self-reported urge to consume high-calorie food items as behavioural marker of the Pavlovian system by computing the average rating across the urge to consume ratings made for each of the ten high-calorie food pictures for each participant and time point. In the next step, we computed the average self-reported valence scores for these stimuli assessed for each participant and time point to determine a behavioral marker of food value. Finally, we computed an impulsivity score as behavioral marker for goal-directed DM for each participant and time point. This was done with the same numerical approach as presented in Weygandt et al. (2015) and based on a hyperbolic discounting model (Mazur, 1987). In particular, this model posits that the value of delayed reward can be computed as value = magnitude/(1 + k · delay). In the framework of the present study, magnitude corresponded to the delayed meal size level of a trial (i.e. two to five), delay assesses the time to gratification for a trial in minutes, and k is a participant-specific marker of impulsivity. To compute behavioural impulsivity, we selected the parameter k from a set of systematically varied parameters k which minimized the sum of squared differences between the true delayed meal value (considered to be 1 for delayed choice trials, and 0 otherwise) and the value predicted by the above equation across all 80 trials included in both delay discounting runs of a participant and time point. After this optimal parameter k was determined, we computed a log-transformation for k, (i.e. $k_{ln} = \ln [k + 0.001]$) to compensate for the skewed distribution of k (Kishinevsky et al., 2012).

2.5. MRI sequences

At T-3, brain images were acquired using a 1.5 T whole-body tomograph (Magnetom Sonata, Siemens, Erlangen, Germany) using a standard 12-channel head coil. In particular, functional images in the cue-reactivity paradigm were measured using a T2*-weighted gradient echo-planar imaging (EPI) blood-oxygenation level dependent (BOLD) sequence comprising 215 images (35 slices, slice thickness = 3 mm; 0.6 mm gap, interleaved; TR = 2000 ms; TA = 57.143 ms; TE = 40 ms; flip angle = 90°; field of view = 192 mm • 192 mm; matrix size = 64 • 64). Brain activity during the delay discounting paradigm was assessed in two runs with the same sequence that, however, included 245 volumes. In both paradigms, the orientation of the axial slices assessed in with these sequences was parallel to the AC-PC line. Moreover, high-resolution T1-weighted anatomical MR images were acquired with a 3-D-magnetization prepared rapid gradient echo (MPRAGE) MRI-sequence (192 slices; slice thickness 1 mm; in-plane voxel resolution 0.5 · 0.5 mm^2; TR = 1970 ms; TE = 3.68 ms; FA = 15°; FOV = 256 · 256 mm^2; matrix size = 512 · 512). Starting with T0, images were acquired with a 3 T whole-body tomograph (Magnetom Trio, Siemens, Erlangen, Germany) also using a standard 12-channel head coil. The change of the scanner was due to organizational reasons. On the 3 T scanner, the cue-reactivity paradigm was again measured with a BOLD sequence including 215 vol (32 slices, slice thickness = 3 mm; 0.75 mm gap, descending; TR = 2000 ms; TA = 62.5 ms; TE = 30 ms; flip angle = 78°; field of view = 192 mm • 192 mm; matrix size = 64 • 64). The delay discounting paradigm was measured with the same sequence that however comprised 245 scans. Finally, also an MPRAGE sequence was acquired (192 slices; slice thickness 1 mm; in-plane voxel resolution 1 · 1 mm^2; TR = 1900 ms; TE = 2.52 ms; FA = 9°; FOV = 256 · 256 mm^2; matrix size = 256 · 256). Specifically, the anatomical MR images were acquired to facilitate the computation of a gray matter (GM) group mask (see below).

2.6. fMRI preprocessing and within-participant brain activity modeling

We conducted standard preprocessing of fMRI data of both paradigms with SPM12 (Wellcome Trust Centre for Neuroimaging, Institute of Neurology, UCL, London UK – http://www.fil.ion.ucl.ac.uk/spm) that included realignment, unwarping, slice time correction, spatial normalization and smoothing with an isotropic Gaussian kernel (8 mm full width at half maximum). A high-pass filter (128 s cut-off) and an autoregessive (AR[1]) model were applied for preprocessing in the temporal domain.

After these basic preprocessing steps, we computed voxel-wise markers of brain activity for both paradigms with multiple regression in first-level general linear model (GLM) SPM analyses. In particular, to compute markers of neural food cue responsivity, we generated boxcar regressors for the four conditions (high- and low-calorie pictures, utensils, and control stimuli) that modeled the sequential block-wise presentation of the ten pictures in a category with a duration of 34.87 s of each block. In the next step, these regressors were convolved with the hemodynamic response function (HRF) and served as covariates of interest. Six motion parameters and a constant were added to the model as covariates of no interest. The resulting regression coefficient voxel-maps for high-calorie food and control pictures entered Analyses 1 and 2. Please note that we focus on high-calorie (and not low-calorie) food pictures because high-calorie foods appear much more potent at altering food reward processing. Specifically, animal studies have shown that access to and consumption of high-calorie foods (and heroin and cocaine) but not low-calorie foods alters (i.e. heightens) reward thresholds in neurophysiological self-stimulation experiments (Kenny, 2011).

To compute neural markers of goal-directed DM for each of the two runs in the delay discounting paradigm, we generated boxcar regressors modeling the 6 s spanned together by the choice and feedback stage for difficult and easy food choice trials separately. After these regressors were convolved with the HRF, the convolved regressors, six motion parameters, and a constant were included in the GLM. Compatible with McClure et al. (2004), and identical to Weygandt et al. (2015), the classification of the 40 trials of a run into 20 difficult and 20 easy ones was performed in a run-wise fashion based on the median food choice or reaction time (RT) respectively. In particular, if the RT of a trial was equal to or larger than the median RT of that run, it was classified as difficult trial and as easy otherwise. Please note, that prior to median determination the RT measures were smoothed in a two-step procedure to improve the signal-to-noise ratio of these measures. Specifically, in the first step, the ten raw RT values belonging to each of the four delayed meal magnitude levels in a run were smoothed separately for each given magnitude level with local regression smoothing. In a second step, the four smoothed RT values belonging to each of the ten delay levels that resulted from the first step were smoothed separately for each given delay level. After the regression coefficient voxel-maps were determined for both difficulty levels and delay discounting runs, we averaged them across runs. These average maps for difficult and easy decisions were then entered into Analysis 3.

2.7. Determination of a gray matter group mask

To constrain all fMRI analyses to brain regions located in GM, we generated a group mask for this tissue type in two steps. Specifically, in the first we computed the voxel-wise average probability of each voxel to belong to gray matter, white matter, and cerebrospinal fluid across all participants and time points (i.e. across 100 data points per voxel). This was done based on unmodulated, spatially normalized tissue probability maps which were computed based on the T1-weighted MPRAGE images with the segmentation algorithm included in SPM12. Voxel coordinates for which the average gray matter probability was larger than the average for the two other tissues were added to the mask. In the second step, we added the six direct voxel neighbors of each voxel coordinate included in step one (i.e. the six voxels for which the Euclidean distance was one voxel) to account for the spatial smoothing of the functional brain images. Finally, those voxel coordinates were entered into Analyses 1–3 that were located in the GM mask and that were covered by all fMRI scans of all participants and time points.

2.8. fMRI analysis

Four longitudinal fMRI analyses were conducted. In Analyses 1–3, we searched for the neural basis of Pavlovian and goal-directed DM mechanisms on the voxel level. In Analysis 4, we proceeded to study the impact of interactions between Pavlovian and goal-directed controllers on dietary variables on a network level. All analyses were performed with LME models which can handle participant-specific signal trends and unbalanced data due to participant drop-out (Verbeke and Molenberghs, 2000) thus allowing for more powerful analyses of longitudinal data (Bernal-Rusiel et al., 2013a; b). In short, LME regression models explain the criterion variance as a linear combination of fixed and random effects. Fixed effects correspond to factors that can be defined by the experimenter (e.g. group membership), random effects to those which can not but that vary in a participant-specific fashion such as participant-specific signal means or trends. In matrix notation, LME models are formulated as $\mathbf{y} = \mathbf{X} \cdot \mathbf{b} + \mathbf{Z} \cdot \mathbf{u} + \mathbf{e}$. Specifically, $\mathbf{y}$ corresponds to the criterion vector, $\mathbf{X}$ ($\mathbf{Z}$) to the design matrix of fixed (random) effects, $\mathbf{b}$ ($\mathbf{u}$) to their estimates, and $\mathbf{e}$ to the unexplained criterion variance. The LME analyses were computed with in-house software (cf. Weygandt et al., 2013, 2012) using the FITLMEMATRIX algorithm included in Matlab 2014a) (MathWorks, Natick, Massachusetts, USA). The software will be made available by the corresponding author on request.

2.8.1. Analysis 1: brain regions showing stable Pavlovian responses across the study period

In this analysis we searched for brain regions where responses to high-caloric food pictures were higher than to neutral control pictures in the cue-reactivity task in a stable fashion across all time points. Analysis 1 was conducted in two steps: In the first ('model selection'), we selected a suitable random effects model for each voxel. This is important because the chosen random effects model can have an impact on the longitudinal data covariance structure which in turn can affect the test statistic (Gurka et al., 2011). In the second ('test'), we added a regressor for the fixed effect of interest to the model chosen in the selection stage and computed the corresponding test statistics reported in the Results section. This separation into model selection and test was performed to avoid that the test statistic could be affected by model selection.

In order to select a suitable random effects model for each voxel in the first step, we compared three different mixed-models for the signal variation in that voxel. This signal variation was summarized by vector $\mathbf{y}$ which contained the voxel regression coefficients computed in the within-participant GLM analysis for high-calorie food and control pictures that were pooled across all participants and time points. Thus, $\mathbf{y}$ contained 200 elements per voxel. The three models uniformly included the same fixed effects: regressors for sex, group (intervention or ad-libitum), age at diet onset, and a constant. Additionally, two time-varying regressors were included, (i.e. one coding for the time after diet onset, and another for the time after diet offset which coded zero for earlier time points). These two latter regressors were needed to capture non-linear effects of only temporary participation in a diet (i.e. from T-3 to T0, not from T0 to T36). This so-called piecewise-linear regression approach is a standard approach for modeling of interrupted time series (Bernal-Rusiel et al., 2013a) typically observed in treatment studies (Wagner et al., 2002). The random effects varied across the three models and either contained only the random intercept, the random intercept plus the time after diet onset, or the random intercept plus the time after diet onset and offset.

After each of the three models was computed for each voxel, a voxel-wise likelihood-ratio test was performed with the COMPARE algorithm implemented in Matlab for all three possible pair-wise model comparisons to evaluate the relative suitability of the different models. The significance of these tests was assessed using a significance threshold corrected for multiple comparisons (i.e. 0.05 divided by the number of voxels included in the gray matter). Based on these three comparative voxel-wise goodness of fit tests, and using a simple random effects selection scheme depicted in Supplementary Table S1, we determined the most appropriate random effects model.

In the test stage, we added a regressor for the fixed effect of interest (i.e. a vector coding ones for high-calorie food picture presentation and zeros for control picture presentation) to the selected model to generate and compute the most appropriate test model. All models in Analysis 1 were computed based on 200 data points (i.e. 2 conditions · [T-3: 22 + T0: 27 + T12: 24 + T24: 17 + T36: 10] participants). Fig. 4 below illustrates the longitudinal voxel-wise LME brain activity modelling approach (for an LME model with the most complex of the above-mentioned random effects models). Please see the Supplement for further details on LME models used in Analysis 1.

In the Results, we report brain areas fulfilling two criteria simultaneously. First, the t-statistic of a given voxel had to reach a significant positive t-statistics for the fixed effect of interest according to a threshold corrected for family-wise error (FWE; $\alpha_{FWE} = 0.05$) on the voxel-level. Specifically, this threshold was computed with the Bonferroni-method by dividing 0.05 by the number of voxels located in the GM mask (approximately 40000). Second, voxels fulfilling the first criterion were only considered meaningful (i.e. were reported) if at least five neighboring voxel coordinates fulfilled the first criterion to get rid of spurious activations. For voxel clusters fulfilling both criteria, we report the cluster size, the coordinates of the voxel with the largest t-statistics in that cluster, the false positive rate, and t-statistic of this voxel. Additionally, we report the effect size parameter f^2 (Cohen, 1988) for this peak voxel which was computed following the heuristic for LME models presented in Selya et al. (2012). According to Cohen (1988), $f^2 \geq 0.02$ indicates a weak, $f^2 \geq 0.15$ a medium, and finally $f^2 \geq 0.35$ a strong effect. Finally, we used an algorithm developed by Bernal-Rusiel et al. (2013a; https://github.com/NeuroStats/lme) to determine the power of

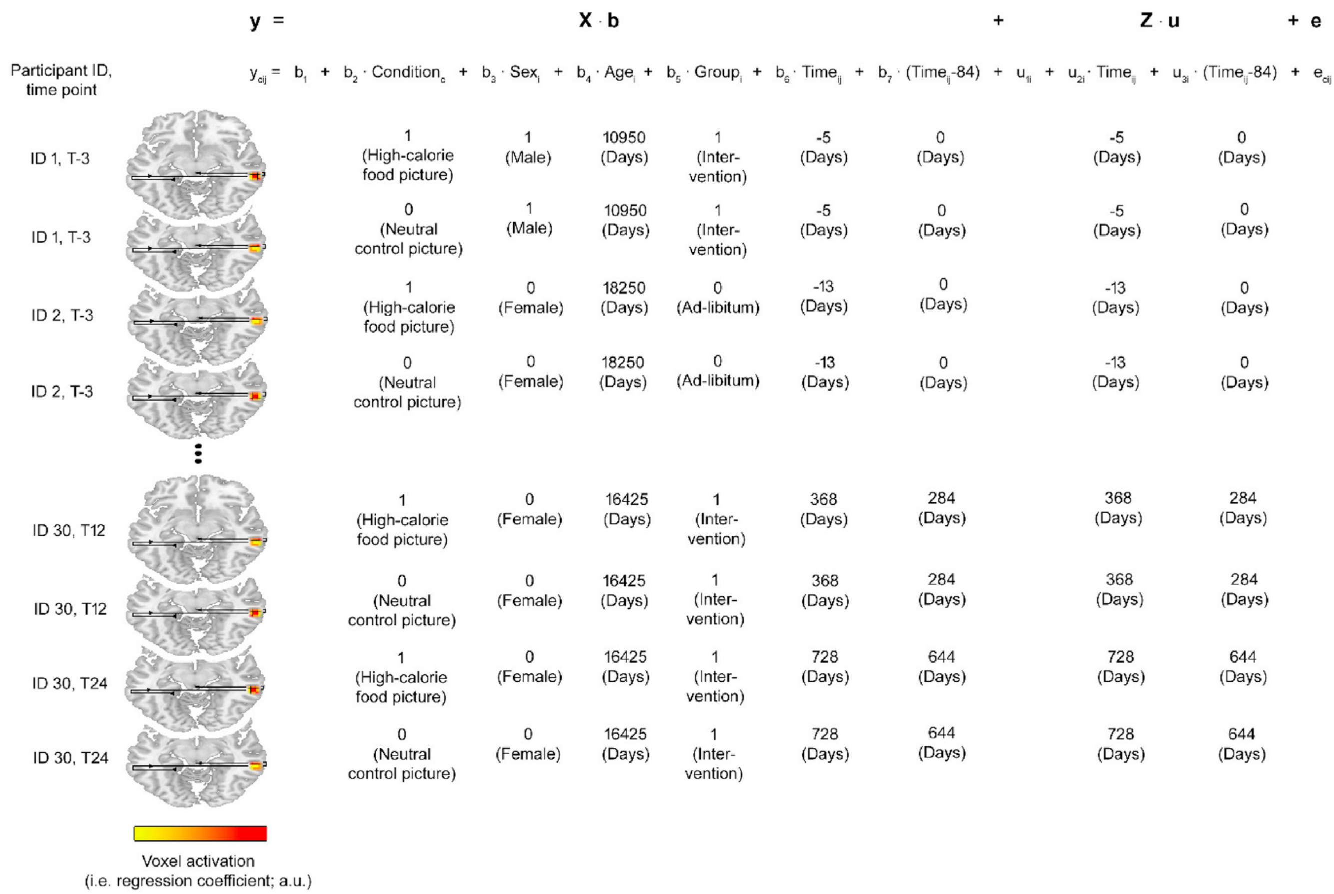

Fig. 4. Shows the longitudinal voxel-wise brain activity modelling approach using LME regression for an exemplary voxel in Analysis 1 (please note, that the same approach was used in Analyses 2 and 3). The equation in the top row shows the LME model computed in matrix notation, the row below in element-wise notation. For the LME model equation in element-wise notation, c indexes condition (1: high-calorie food picture; 0: neutral control picture), i the participant, and j the time point. Sex_i: participant's sex = 1 if male, 0 female. Age_i: is the age at diet onset in days. $Group_i$: participant's group = 1 if intervention, 0 ad-libitum. $Time_{ij}$: time after diet onset in days. $Time_{ij}$-84 corresponds to the time after diet offset and is $Time_{ij}$-84 if $Time_{ij} > 84$ and 0 otherwise. y_{cij} corresponds to a regression coefficient of a voxel determined in within-participant brain activity modelling for high-calorie food and control picture presentation for a participant and time point.

tests conducted at significant peak voxel coordinates in Analysis 1. In this voxel-wise power or sensitivity estimation procedure respectively, the abovementioned α_{FWE} parameter was used as false positive rate. Please note, that we also present voxel-wise sensitivity parameters for all tests conducted in Analysis 1 (and 2 and 3) irrespective of whether these tests reached significance in supplementary fMRI analysis S1. Moreover, in supplementary fMRI analysis S2 we evaluated a potential impact of the MRI scanner change between T-3 and T0 on the neural substrates of decision-making identified in Analyses 1 (and 2 and 3), by repeating these analyses based on fMRI scans acquired in the period from T0 to T36.

2.8.2. Analysis 2: brain regions showing a decline of Pavlovian responses across the study period

Given that a weight-reduction diet can be seen as Pavlovian extinction learning because food cues become less frequently coupled to reward (e.g. Peters et al., 2009), we searched for areas showing a significant negative effect of time after diet onset on brain activity in the cue-reactivity task in this analysis. Findings reported for Analysis 2 were not determined by individual LME models but instead correspond to the results obtained by the LME test models computed in Analysis 1 for the fixed effect regressor 'time after diet onset'. All other aspects were as reported for Analysis 1.

2.8.3. Analysis 3: neural substrates of goal-directed control

In this analysis, we searched for brain regions responding stronger during difficult than easy food decisions across all participants and time points. Except for the fact that regression coefficient maps evaluated in Analysis 3 reflected neural responses during difficult and easy food decisions and the smaller number of data points available (i.e. 78 data sets times two conditions = 156 data points; cf. Fig. 1), all other aspects were identical to Analysis 1. Please note that the number of data sets available for the analysis of goal-directed decision-making in Analysis 3 was smaller than for the analysis of Pavlovian responses (Analyses 1 & 2) since some participants showed invariant decision-making behavior (i.e. they exclusively chose either the immediate or the delayed option across the whole task). When this was the case, the data of that participant were excluded from Analysis 3 for a given time point since it was impossible to compute behavioral impulsivity scores under these circumstances. Additionally, participants that differed in terms of decision-making behavior between a behavioral pilot and the fMRI delay discounting task conducted at their first visit were excluded. Specifically, the data of 6 participants were excluded due to inconsistent or invariant decision-making behavior at T-3 and the data of 4/5/5/2 participants were excluded due to these reasons at T0/T12/T24 or T36 respectively.

2.8.4. Analysis 4: prediction of dietary variables based on decision-making network parameters

Having now defined the neural basis of DM mechanisms on a voxel-level, we proceeded to study the interaction between areas with stable Pavlovian responses and goal-directed controllers as well as between regions with declining Pavlovian responses and goal-directed control areas on a network-level. Specifically, we were interested in the degree to which interactions between these two pairs of systems predicted the following three dietary variables: (i) future dietary success ('ΔBMI'), (ii) urges to consume high-calorie foods, and (iii) impulsivity. In addition, we also tested whether the individual DM systems contain predictive information for the three dietary variables to evaluate the importance of DM system interactions and to assess the overlap of our results with results of existing single DM system fMRI studies (e.g. Weygandt et al., 2015, 2013; Murdaugh et al., 2012; Hare et al., 2009; McClure et al., 2004).

Neural network parameters characterizing the interaction between the two pairs of DM systems as well as the functioning of the individual DM systems were computed in a three-step procedure based on brain activity at regions identified in Analyses 1 to 3. Specifically, in the first step we determined the differences between neural responses to high-calorie food minus control pictures for each participant and time point at all peak voxel coordinates identified in Analysis 1 (three peak coordinates) and 2 (eleven peak coordinates) and included these difference data in one matrix per Analysis. The same was done for neural responses to difficult and easy food decisions at all seven peak coordinates found in Analysis 3. In the second step, we computed DM network activity parameters for the individual DM systems with Principal Component Analysis (PCA), an established technique in neuroimaging for the analysis of neural network behavior (cf. Bullmore et al., 1996). In particular, PCA was used to determine all principal components (PCs) for the voxel response difference signals computed in the first step for each of the three voxel-wise analyses separately and thus yielded three PCs for Analysis 1, eleven PCs for Analysis 2, and seven PCs for Analysis 3. Please note that we primarily decided to use PCs instead of manifest longitudinal voxel signals for the prediction of dietary variables in Analysis 4 because the voxel signals at peak coordinates identified in Analyses 1 to 3 were strongly influenced by the participants' signal means. Consequently, also the mutual correlation of signals was strong for the peak voxels found in each voxel-wise analysis. Principal components, however, are mutually orthogonal (Jolliffe, 1986) and thus facilitate a better utilization of the heterogeneity of Pavlovian and goal-directed signals for prediction of dietary variables. Finally, in the third step, network interaction parameters for the two pairs of systems were computed as the element-wise product of each possible pair of PCs for a given pair of systems (i.e. 21 and 77 interaction parameters for the two pairs of systems).

After the PCs of the individual DM systems and the interaction parameters were computed, each of these parameters was separately added as fixed effect of interest to the design matrix **X** of one LME model used to predict ΔBMI, to another model used to predict urges to consume high-calorie foods, and finally to another one used to predict impulsivity. In analyses of interactions between pairs of systems, the two PCs taken from the individual DM systems to compute a given interaction regressor were additionally included in **X** as fixed effects of no interest to guarantee that identified interaction effects do not reflect their main effects. For prediction of all three dietary variables, **X** also included participants' sex, age at diet onset, group membership, time after diet on- and offset and a constant as fixed effect of no interest. For the prediction of ΔBMI, the BMI score of a participant was included in **X** additionally. In particular, for each ΔBMI-score computed as the BMI difference of a later minus the preceding time point, the BMI of the preceding time point was included at the corresponding position (i.e. row) of **X**. Finally, a participant-specific constant was included in the random effects design matrix **Z** to capture the mean of the criterion variable for each participant. The equations below depict the LME models computed for prediction of ΔBMI based on activity of individual DM systems and based on interactions between DM systems. Models used for predicting urges to consume high-calorie foods and behavioral impulsivity are shown in the Supplement.

$$\Delta BMI_{ij} = b_1 + b_2 \cdot PC_{ij} + b_3 \cdot BMI_{ij} + b_4 \cdot Sex_i + b_5 \cdot Age_i + b_6 \cdot Group_i + b_7 \cdot Time_{ij} + b_8 \cdot (Time_{ij}\text{-}84) + u_{1i} + e_{ij}$$

The above LME model equation was used to predict longitudinal variations in body mass across participants and time based on activity of individual DM systems. In this equation PC_{ij} corresponds to a given component score for either a Pavlovian or the goal-directed DM network of a participant and time-point respectively. BMI_{ij} corresponds to the body mass index of a participant and time-point. The meaning of the other variables is identical to the ones presented in Fig. 4.

$$\Delta BMI_{ij} = b_1 + b_2 \cdot (PC_{Aij} \cdot PC_{Bij}) + b_3 \cdot PC_{Aij} + b_4 \cdot PC_{Bij} + b_5 \cdot BMI_{ij} + b_6 \cdot Sex_i + b_7 \cdot Age_i + b_8 \cdot Group_i + b_9 \cdot Time_{ij} + b_{10} \cdot (Time_{ij}\text{-}84) + u_{1i} + e_{ij}$$

This LME model was used to predict longitudinal changes in body mass across participants and time points based on interactions between DM systems. PC_{Aij} corresponds to the component score of a Pavlovian system for a participant and time-point and PC_{Bij} corresponds to the corresponding goal-directed PC score.

Fifty-two data points were available for ΔBMI modelling (and modelling of all other subanalyses conducted; cf. Fig. 1) in Analysis 4. This smaller number (compared to Analyses 1–3) followed from three aspects. First, only one diet-related parameter was included in **y** for each participant and time point instead of the two parameters in Analyses 1–3 (one voxel regression coefficient for each of the two conditions evaluated in both tasks). Second, computation of a ΔBMI-score for a participant and time point necessitated the availability of BMI scores for two time points (a later and a preceding earlier time point). Given that we aimed to predict future weight changes from the perspective of each ('earlier') time point, only parameters assessed from T-3 to T24 could be used as components of **X** or **Z**. Third, since we wanted to relate Pavlovian and goal-directed decision-making signals in the interaction analyses, we could only evaluate as much data points as were available for the smaller delay discounting data set.

We report positive and negative effects significant on an $\alpha_{FWE} = 0.05$ level. The significance threshold was computed by dividing 0.05 by the number of tests conducted (interaction stable Pavlovian and goal-directed system: 21; interaction declining Pavlovian and goal-directed system: 77; stable Pavlovian responses: 3; declining Pavlovian responses: 11; goal-directed system: 7). Finally, we report the effect size f^2 (Cohen, 1988) and the sensitivity for each test conducted in Analysis 4. The FWE-corrected false positive rate for sensitivity estimation was calculated as 0.1 divided by the number of tests conducted in each analysis to account for the fact that we tested positive as well as negative effects on an $\alpha_{FWE} = 0.05$ level. Please note that we also present sensitivity parameters for all tests conducted in Analysis 4 irrespective of whether they reached significance in supplementary fMRI analysis S1. Furthermore, we also evaluated a potential impact of the MRI scanner change between T-3 and T0 on predictability of dietary variables in supplementary fMRI analysis S3. Finally, we conducted supplementary behavioral analyses testing the mutual links between the dietary variables, their link with ΔBMI, and whether the urges of consumption marker reflected different processes at different levels of food consumption (i.e. before and after the diet).

3. Results

3.1. Sample characteristics

Table 1 below depicts demographic and diet-related sample characteristics separately for each fMRI measurement time point.

Across all 22 participants included in the study at T-3, the measurements for this time point were made on average M = 10 (SD: 23; R: 19 - 89) days prior to diet onset. Eight of the 22 participants were measured slightly (i.e. M = 8.1; SD = 7.1; R: 1–19 days) after diet onset due to organizational reasons such as limited scanner or physician availability. Moreover, across all 27 participants included at T0, data for this time point were on average acquired M = 94 (SD: 16; R: 64–131) days after onset of the diet (which had a duration of 84 days). In 5 of these 27 participants, the T0 measurement was made before (M = 14.8; SD = 5.2;

Table 1
Abbreviations: a.u., arbitrary units.

	T-3	T0	T12	T24	T36
Age (yrs.)	M = 48.4 SD = 14.1 R: 23.5–68.1	M = 46.5 SD = 13.5 R: 22.6–68.1	M = 48.1 SD = 14.4 R: 23.5–69.7	M = 47.6 SD = 14.3 R: 24.8–68.8	M = 40.5 SD = 12.3 R: 25.4–64.6
Sex (male/female)	5/17	8/19	7/24	7/10	3/10
Group (intervention/ad-libitum)	9/13	12/15	11/13	8/9	4/6
Urge to consume (a.u.)	M = 0.4 SD = 0.2 R: 0.0–0.7	M = 0.3 SD = 0.2 R: 0.0–0.7	M = 0.3 SD = 0.2 R: 0.0–0.9	M = 0.4 SD = 0.2 R: 0.0–0.9	M = 0.4 SD = 0.2 R: 0.1–0.9
Valence (a.u.)	M = 4.5 SD = 1.4 R: 2.3–7.4	M = 4.7 SD = 1.6 R: 1.4–8.7	M = 5.2 SD = 1.5 R: 2.8–8.6	M = 5.2 SD = 1.4 R: 1.7–7.9	M = 5.1 SD = 1.4 R: 3.1–7.3
Impulsivity (ln [k + 0.001])	M = −1.3 SD = 0.6 R: −2.4 - 0.19	M = −1.1 SD = 0.9 R: −2.5 - 0.9	M = −0.9 SD = 1.1 R: −2.5 - 1.5	M = −1.4 SD = 0.8 R: −2.5 to −0.1	M = −1.5 SD = 0.7 R: −2.5–0.4
BMI (kg/m^2)	M = 34.4 SD = 3.1 R: 29.6–41.8	M = 32.0 SD = 3.9 R: 23.3–42.7	M = 32.3 SD = 3.7 R: 23.4–40.6	M = 33.1 SD = 4.2 R: 23.6–42.5	M = 33.7 SD = 5.5 R: 23.8–43.2

	T-3 → T0	**T0 → T12**	**T12 → T24**	**T24 → T36**
ΔBMI (kg/m^2)	M = −4.5 SD = 1.8 R: −7.8 to −1.6	M = 1.3 SD = 1.5 R: −1.4 - 3.6	M = 0.7 SD = 1.7 R: −1.4 - 3.6	M = 0.7 SD = 2.3 R: −3.2 - 3.8

R: 8–20 days) the end of the diet for organizational reasons. On average the MRI measurement for T12 took place M = 479 (SD: 57; R: 387–593) days after diet onset, 875 (SD: 54; R: 784–1019) days at T24, and finally 1140 (SD: 51; R: 1086–1250) days after diet onset at T36.

3.2. fMRI results

Three analyses were conducted to identify the neural basis of DM mechanisms on the voxel-level. Specifically, Analysis 1 searched for areas with stable Pavlovian responses and revealed three voxel clusters whose peak coordinates were located in middle and inferior occipital gyri and spread to fusiform gyrus and temporal cortex (Table 2a, Fig. 5a). Analysis 2 searched for regions showing declining Pavlovian responses across time and found eleven voxel clusters whose peak coordinates were located in (medial) superior frontal gyrus, middle frontal gyrus, the orbital part of the inferior frontal gyrus, anterior and midcingulate cortex, and hippocampus (Table 2b, Fig. 5b). In Analysis 3, we searched for neural correlates of goal-directed control which revealed seven voxel clusters whose peak coordinates were centered in (pre-) supplementary motor area (SMA), precentral gyrus, insula, opercular part of inferior frontal gyrus, and inferior parietal gyrus. In addition to these locations of peak coordinates, the clusters also extended into angular gyrus, anterior and midcingulate cortex, triangular and orbital inferior frontal gyrus, and middle, superior and medial superior frontal gyri (Table 2c, Fig. 5c).

In Analysis 4, we evaluated the importance of interactions between both Pavlovian systems and goal-directed controllers for future dietary success (ΔBMI) and behavioral dietary factors with longitudinal network analyses. This revealed a strong negative association between interactions of stable Pavlovian and goal-directed DM signals ($t = -4.7$, $p < 10^{-4}$, $f^2 = 0.59$) and the ΔBMI. PC signals computed for goal-directed control that contributed maximally to this interaction were most closely related to voxel signals in the right insular peak coordinate.

Table 2
Brain activity related to Pavlovian and goal-directed DM systems as revealed by Analyses 1 to 3. Abbrev.: MNI, Montreal Neurological Institute; BA, Brodmann area; CS, cluster size in mm^3; x, y, z, coordinates in MNI space. t, t-statistic; p_{FWE}, family-wise-error corrected probability for observing the given t-value by chance on the voxel level; f^2, Effect size; SEN, sensitivity; inf., inferior; gy., gyrus; mid., middle; sup., superior; pt., part.

Region MNI	BA	CS	x	y	z	t	p	f^2	SEN
a) Areas with stable Pavlovian responses									
Inf. occipital gy.	37	4860	45	−61	−13	8.3	$<10^{-13}$	0.50	1.00
Inf. occipital gy.	19	5373	−45	−73	−7	7.3	$<10^{-11}$	0.40	0.98
Mid. occipital gy.	18	621	24	−88	11	6.2	$<10^{-8}$	0.28	0.86
b) Areas with declining Pavlovian responses									
Inf. frontal, orbital pt.	47	567	42	35	−10	−8.5	$<10^{-14}$	0.76	1.00
Sup. frontal gy.	32	702	−15	53	23	−8.0	$<10^{-12}$	0.59	1.00
Hippocampus	37	135	−39	−37	−7	−7.2	$<10^{-10}$	0.48	0.98
Anterior cingulate cortex	32	459	9	47	23	−7.2	$<10^{-10}$	0.48	0.98
Midcingulate cortex	23	216	12	−46	32	−7.0	$<10^{-10}$	0.40	0.97
Inf. frontal, orbital pt.	47	324	−33	29	−16	−6.8	$<10^{-10}$	0.48	0.98
Mid. frontal gy.	8	459	30	23	53	−6.5	$<10^{-9}$	0.42	0.94
Anterior Cingulate Cortex	11	567	−15	41	−1	−6.5	$<10^{-9}$	0.33	0.89
Sup. frontal gy., medial part	10	135	3	59	17	−6.3	$<10^{-8}$	0.38	0.92
Sup. frontal gy., medial part	10	270	−18	59	8	−6.2	$<10^{-8}$	0.25	0.87
Mid. frontal gy.	46	135	21	50	11	−6.0	$<10^{-8}$	0.32	0.83
c) Goal-directed controllers									
Supplementary motor area	32	2079	−12	14	47	7.4	$<10^{-11}$	0.65	0.98
Precentral gy.	6	2079	−48	2	38	6.5	$<10^{-9}$	0.42	0.90
Insula	47	837	30	26	−4	6.4	$<10^{-8}$	0.38	0.86
Insula	47	297	−36	23	−4	6.1	$<10^{-8}$	0.36	0.82
Inf. frontal, opercular pt.	44	702	36	11	35	5.8	$<10^{-7}$	0.34	0.73
Inf. parietal gy.	40	783	42	−52	53	5.5	$<10^{-7}$	0.24	0.67
Supplementary motor area	46	243	3	8	53	5.2	$<10^{-6}$	0.29	0.54

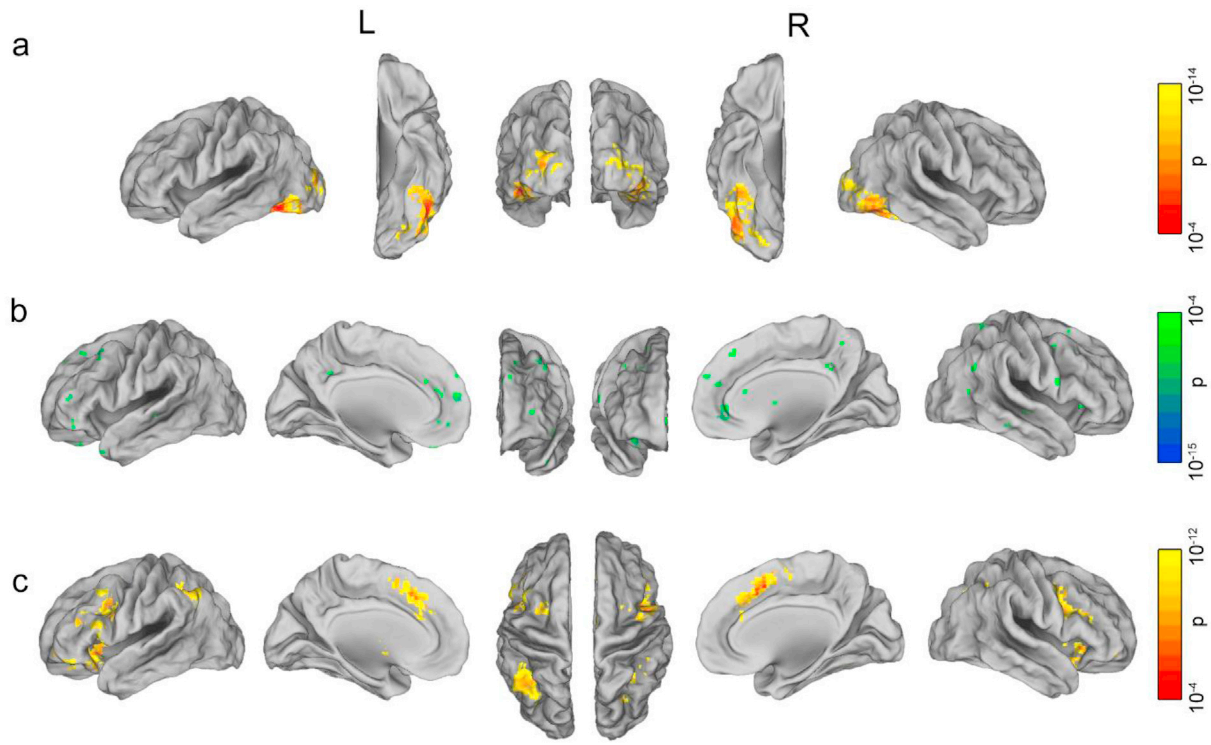

Fig. 5. Illustrates regions with a) stable Pavlovian responses across the study period, b) a decline in neural responses in the cue-reactivity task across the 39-month, and c) stronger responses during difficult compared to easy food choices across the full period. For visualization purposes, we thresholded the statistical maps at $p = 10^{-4}$ (uncorrected). Please note that the cortical surface representations in this figure were generated by mapping the corresponding false positive rates computed in voxel-space onto a three-dimensional cortical surface model using Caret (Van Essen, 2012). Abbrev.: L, left, R, right.

Predetermined by the fact that all peak voxel coordinates found in Analysis 1 were located in visual areas, stable Pavlovian PC signals contributing maximally to this interaction were also related to voxel signals in these areas. The subanalyses included in Analysis 4 aiming to evaluate whether PC signals of the individual DM systems contain predictive information for the three dietary variables found a moderate positive link of declining Pavlovian signals primarily in left hippocampus with ΔBMI ($t = 3.1$, $p = 0.002$, $f^2 = 0.22$; i.e. a negative link with dietary success). Moreover, a moderate negative association of goal-directed DM activity primarily in right inferior parietal gyrus and ΔBMI was found ($t = -2.7$, $p = 0.005$, $f^2 = 0.16$). Finally, Analysis 4 revealed a strong positive association between stable Pavlovian response activity (centered in visual areas) and urges to consume high-calorie foods ($t = 2.6$, $p = 0.007$, $f^2 = 0.41$). No association was found between neural DM signals and behavioral impulsivity. See Fig. 6 for details.

4. Discussion

In this study, we identified the neural basis of Pavlovian and goal-directed decision-making systems and evaluated the importance of interactions between these systems for future three-year dietary treatment success obtained by individuals with obesity. Four fMRI analyses were conducted. In the first, we searched for Pavlovian response areas showing a stronger activity during presentation of high-calorie food than control pictures in a stable fashion across the full study period and found three voxel clusters centered in occipital cortex and extending into fusiform gyrus and temporal cortex. These findings are in line with research on responsivity to reward-cues in healthy (e.g. Bradley et al., 2003) and persons with obesity (Martin et al., 2010; Rothemund et al., 2007). For example, Bradley et al. (2003) found in a cue-reactivity study contrasting reward-related images to control pictures that large portions of the visual cortex respond stronger to the non-control pictures. Moreover, a link between functioning of visual regions and dopaminergic reward areas was demonstrated by Rypma et al. (2015) who showed in a combined fMRI and Positron-Emission-Tomography study that activity in fusiform gyrus measured in a face perception task depends on its dopamine binding potential. Consequently, this suggests that the hyperresponsivity of visual areas to food stimuli that was found in Analysis 1 across the full study period reflects more extensive visual processing of reward-related than neutral control stimuli.

Nevertheless, although these areas are thus connected to reward regions, they are not the Pavlovian control regions primarily assumed to mediate the association of initially neutral stimuli (food pictures) and reinforcers that can trigger consumption (e.g. striatum, prefrontal cortex, hippocampus; Volkow et al., 2017; Rangel, 2013; Robbins et al., 2008; Ito et al., 2005). Although this might be surprising on first sight, it might become understandable when considering that a weight-reduction diet can be interpreted as Pavlovian extinction learning since participation in such a diet reduces the coupling of food cues and reward (e.g. Peters et al., 2009). In line with this view, we tested in Analysis 2 whether the weight reduction might have altered neural responsivity to food cues by searching for areas showing a linear response decline across the study period. This revealed coordinates in (medial) superior and middle frontal gyrus, orbital part of the inferior frontal gyrus, anterior and midcingulate cortex, and hippocampus. In addition, a striatal area (MNI: −24, −13, 23; $t = -6.0$; $p < 10^{-8}$; volume = 81 mm^3) showed such a signal decline on an $\alpha_{FWE} = 0.1$ level. Together, these areas correspond very well to the regions typically assumed to mediate Pavlovian control (Volkow et al., 2017; Rangel, 2013; Robbins et al., 2008; Ito et al., 2005).

An influential model on aspects of reinforcer evaluation (Berridge et al., 2009) might help to interpret findings made in Analyses 1 and 2. Specifically, this model posits that (i) this evaluation comprises two components, a motivational ‘wanting’ and a value-related ‘liking’ component, (ii) both parameters are tightly coupled during physiological levels of consumption, and (iii) wanting increases in periods of overconsumption whereas liking decreases in the same time. Thus, when

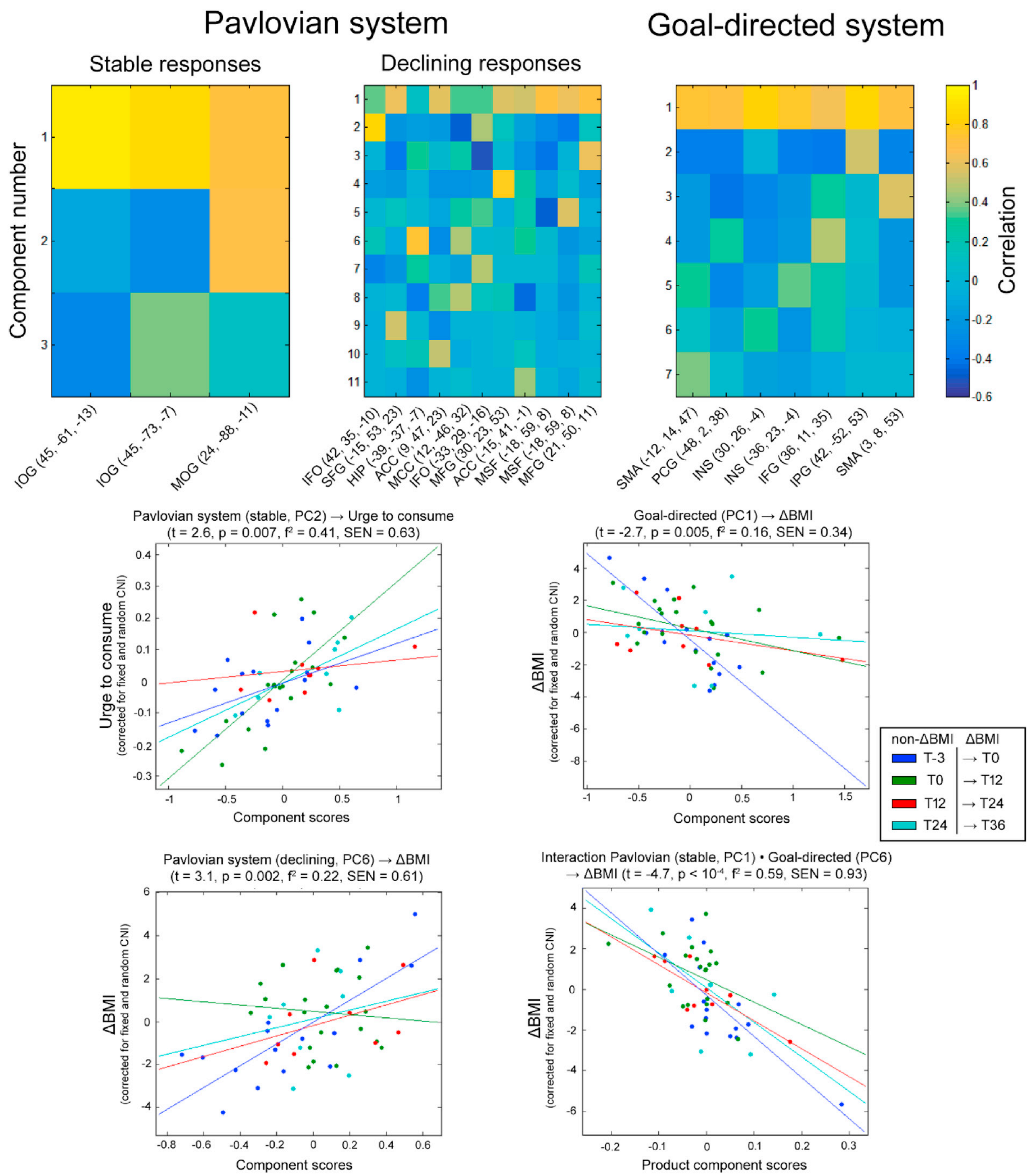

Fig. 6. Longitudinal prediction of dietary variables based on neural DM signals. The three images in the top row depict the correlations of manifest voxel signals extracted from coordinates identified in Analysis 1–3 and the corresponding PCs computed in Analysis 4. The scatterplots depict the results of Analysis 4 for the prediction of ΔBMI and the urge to consume high-calorie food pictures based on PCs determined for the individual DM systems and their interactions. Abbrev.: ACC, anterior cingulate gyrus; CNI, covariates of no interest; f^2, effect size; HIP, hippocampus; IFG, inferior frontal gyrus; IFO, inferior frontal gyrus, orbital part; INS, insula; IOG, inferior occipital gyrus; IPG, inferior parietal gyrus; MCC, midcinuglate gyrus; MFG, middle frontal gyrus; MOG, middle occipital gyrus; MSF, medial superior frontal gyrus; PC, principal component; PCG, precentral gyrus; SEN, sensitivity; SFG, superior frontal gyrus; SMA, supplementary motor area.

assuming that a substantial part of measurements in our study was made during periods of physiological levels of consumption (after a diet), one could derive from this model that areas identified in Analysis 1 primarily reflect the value aspect, whereas areas found in Analysis 2 relate to the motivational component. These assumptions were also supported by three supplementary behavioral analyses that found that (i) self-reported urges to consume high-calorie foods are positively coupled to future dietary success, (ii) and to the self-reported valence of these stimuli. Moreover, (iii) the variance of the valence parameter that was explained by the urges to consume marker was significantly smaller before the diet than during later periods. See Supplement for further discussions.

In Analysis 3, we investigated neural substrates of goal-directed DM and found several regions that responded stronger during difficult than easy food decisions that were centered in (pre-) SMA, insula, inferior frontal and inferior parietal cortex. Additionally, a large voxel cluster was found whose peak coordinate was located in precentral gyrus but that also spread into middle frontal gyrus or DLPFC respectively. Importantly, this pattern of areas strongly overlaps with areas assumed to evaluate, modulate and implement VMPFC value signals contributing to goal-directed DM according to a neuroeconomic model of dietary DM (Rangel, 2013). In particular, this model assumes that VMPFC first computes value signals for different food choices at hand. If the value of several options is very disparate, choice is easy and the decision can be directly derived from a simple comparison of these value signals without further finetuning. However, given that VMPFC has a tendency to overestimate the value of immediately rewarding options, modulation of VMPFC signals is necessary in case of difficult options having similar values. At this point, DLPFC, inferior parietal cortex, pre-SMA, and precentral gyrus come into play as they can not only compare but also modulate and implement the VMPFC signals. Thus, together they facilitate appropriate dietary DM in difficult choices. Finally, the model assumes that insular cortex contributes to goal-directed dietary DM by updating the value of food after changes of physiological states (e.g. from hunger to satiety).

In Analysis 4, we addressed the main research question of the study and tested whether interactions between Pavlovian and goal-directed DM systems contain predictive information for dietary variables. Using longitudinal fMRI network analyses, we found that an interplay between areas with stable Pavlovian responses and goal-directed DM controllers was positively (negatively) linked to future dietary treatment success (ΔBMI) in obesity. The stable Pavlovian signals were related to activity in visual areas. Their goal-directed counterparts were primarily related to insular activity. Based on these findings and the insular role in goal-directed control (Rangel, 2013), we conclude that adaptation of visual food cue processing resources to goal-directed insular control signals is a prerequisite for sustained dietary success as the latter signals presumably modulate Pavlovian urges triggered by frequent cue exposure in everyday life. On the contrary, interactions of regions with declining Pavlovian responses and goal-directed controllers were not related to dietary variables. Thus, when integrating these findings and the assumptions of Berridge et al. (2009), one might conclude that modulation of food value by the goal-directed system is more important for long-term dietary success than modulation of dysregulated motivational aspects due to the more limited duration of the latter.

To evaluate the relative importance of interactions between DM systems as determinants of future dietary success, we computed the strength of associations between the latter factor and activity of the individual DM systems as references in Analysis 4. This revealed a moderate negative association with activity in areas showing declining Pavlovian responses primarily in left hippocampus. This is well in line with animal studies showing a role of the hippocampus in appetitive Pavlovian conditioning where locomotor activity is coupled to contextual stimuli of reward (e.g. Ito et al., 2005). Moreover, a moderate positive association of dietary success with goal-directed signals primarily located in right inferior parietal gyrus was found. This result is consistent with McClure et al. (2004), who showed that the relative involvement of this region during intertemporal choice is positively associated with more controlled food choices. Taken together, when comparing the association strength obtained between DM system interactions and dietary success to those obtained between individual DM systems and dietary success, one can conclude that system interactions are a more important predictor and should thus not be ignored in future studies.

Despite these novel insights provided by our study in the importance of the interplay between neural DM systems for long-term dietary success, several aspects warrant further discussions. First of all, we want to highlight that we did not explicitly address potential differences between the intervention and the ad-libitum group in this study. We decided not to do so, because (i) we were primarily interested in identifying neural predictors of long-term dietary success, (ii) group membership was included as fixed effect of no interest in all analyses conducted, and (iii) this work is already quite comprehensive. Nevertheless, the impact of endogenous treatment effects on neural decision-making mechanisms might be evaluated more explicitly in future work.

Furthermore, we want to mention that the fasting duration prior to the fMRI measurements varied across participants to a small extent (i.e. participants fasted between 12 and 15 h) due to variations in MR scanner availability. This might have had an effect on the behavioral ratings regarding the high-calorie food items. However, given that the urges to consume high-calorie foods ratings were predicted in Analysis 4 based on the activity of areas identified in Analysis 1 and given that this predictability is perfectly in line with assumptions of increased food wanting in obesity (cf. Castro and Berridge, 2014; Berridge et al., 2009), we do not believe that these small variations hamper the interpretability of our results.

Another question that needs to be addressed is that of the relation of findings made in Weygandt et al. (2015, 2013) and in this study given that both fMRI tasks included in this study comprise the presentation of visual food stimuli. Moreover, results appear to overlap to a certain extent in terms of the larger brain areas found in Weygandt et al. (2013) and in the present study (e.g. DLPFC, anterior cingulate cortex). Despite these apparent similarities, however, we do not believe that the findings made in Weygandt et al. (2015, 2013) anticipate the results of this study primarily due to the difference between processes involved in both fMRI tasks. In particular, the delay discounting paradigm investigated in an isolated fashion in Weygandt et al. (2015, 2013) is a task requiring the participants to engage actively in a set of complex behaviors. For example, participants have to determine and represent the value of different choices, compare these values, initiate and execute motor actions to realize the decision and finally to update value depending on action outcome and in internal states. On the contrary, the cue-reactivity task additionally evaluated in this study triggers neurobehavioral responses in an automatic and passive fashion. Behaviors controlled by this system are simple, automatic, pre-programmed and hardwired behaviors such as the salivation response of a dog when confronted with food (Rangel, 2013). Importantly, this idea (i.e. that the key processes involved in Pavlovian and goal-directed control are largely independent) is supported by the fact that the specific voxel clusters identified in Analyses 1 to 3 did not overlap.

Moreover, the question of whether brain activity processes in regions identified across analyses could be the consequence of weight loss should be discussed. Given that areas identified in Analyses 1 and 3 showed stronger responses to high-calorie food than control stimuli or difficult than easy food choices respectively in a stable fashion across the whole study period despite the fact that BMI varied strongly in the same time makes this very unlikely for these analyses. On the contrary, we assume that to be the case for regions found in Analysis 2 as this analysis explicitly searched for areas with a neural response decline over time to test the assumption that a weight loss diet functions as Pavlovian extinction learning that reduces the coupling between food cues and reward (e.g. Peters et al., 2009). Importantly, however, we would like to emphasize at this point that findings revealed by the key analysis aiming to predict future long-term dietary success (i.e. ΔBMI) based on brain activity (Analysis 4) do not reflect the variation in body mass since the BMI was included in the corresponding LME models as fixed effect of no interest.

Another aspect that should be addressed is the comparably limited number of data points available for some participants which might speculatively impede proper longitudinal data modelling. However, simulation analyses conducted by Bernal-Rusiel et al. (2013a) have shown that the accuracy of LME models that include participants for whom only a single data point is available (i.e. data containing no longitudinal information at all) is higher than of comparison models excluding these participants. Consequently, the inclusion of participants appears justified independent of the extent and sequence of longitudinal data availability (see also Fitzmaurice et al., (2011)).

Partially related to the above point, the statistical power of tests conducted should be discussed. Specifically, retrospective sensitivity analyses (Supplementary fMRI analysis S1) performed to evaluate the power of voxel-wise tests in Analyses 1 to 3 showed that sensitivity was high and sufficient despite the very conservative false positive rate applied in sensitivity estimation to account for the high number of voxel-wise tests. This notion is supported by the fact that the overlap of brain areas identified in our analyses and areas typically found in related cue-reactivity (e.g. Murdaugh et al., 2012; Weygandt et al., 2012; Stice et al., 2010; Martin et al., 2010; Stoeckel et al., 2008; Rothemund et al., 2007) or delay-discounting studies (e.g. Rangel, 2013; Kishinevsky et al., 2012; McClure et al., 2004) was high. Most importantly, the statistical power of tests conducted to address the key question of whether future long-term dietary success of persons with obesity can be predicted based on interactions between Pavlovian and goal-directed decision-making systems in Analysis 4 was also sufficient and high. Nevertheless, it has also to be mentioned at this point that although Analysis 4 found some significant associations between activity of neural decision-making systems and diet-related behavioral variables, the overall statistical power of tests investigating associations between these factors was small to moderate. Consequently, future studies might test such associations based on larger samples.

An important aspect that should be discussed is the question of whether the MRI scanner change between T-3 and T0 had an important impact on the neural substrates of decision-making and the predictability of dietary variables. To address this point, two supplementary fMRI analyses were conducted. In particular, in supplementary fMRI analysis S2, we repeated Analysis 1 to 3 from the main text but now only used fMRI scans acquired in the period from T0 to T36 (which were all assessed with the same MRI scanner). Coordinates revealed by this supplementary analysis strongly overlapped with coordinates identified in Analysis 1 and 3 in the main text (i.e. 98% of voxel coordinates identified in Analysis 1 and 64% of coordinates identified in Analysis 3 were also found in supplementary fMRI analysis S2). Please note that no areas were found in supplementary fMRI analysis 2 that showed a significant response decline over time which is however not surprising given the participants' BMI development after the cessation of the diet. Finally, in supplementary fMRI analysis S3, we repeated Analysis 4 from the main text based on fMRI scans acquired in the period from T0 to T36 to test the potential importance of the MRI scanner change on predictability of dietary variables. This supplementary analysis revealed a considerable overlap with findings obtained in Analysis 4 in two important aspects. First, the strongest link between dietary success and neural decision-making signals (being significant on an uncorrected threshold and moderate to strong in terms of effect size according to Cohen [1988]) was also found for interactions of stable Pavlovian brain activity in the visual system and insular activity related to goal-directed decision-making. Second, supplementary fMRI analysis S3 also found a positive and highly significant link between neural decision-making signals and self-reported urges to consume high-calorie foods. Together, these findings suggest that the change of the MRI scanner between T-3 and T0 did not have an important impact on the neural substrates of decision-making identified and the predictability of dietary variables observed in our study.

A further point that should be mentioned is that food and control stimuli were not explicitly matched for visual characteristics such as image frequency or color content and food pictures were not controlled for portion size. Thus, one might speculate that between category differences in these features might have contributed to identification of areas in Analysis 1 (that were centered in occipital cortex and extended into fusiform gyrus and temporal cortex). Importantly, however, visual areas such as fusiform gyrus are found consistently across fMRI obesity studies applying cue-reactivity paradigms (e.g. Pursey et al., 2014; Murdaugh et al., 2012; Rothemund et al., 2007) which includes studies that explicitly matched food and control stimuli for brightness, resolution, and size (e.g. Martin et al., 2010). Moreover, fusiform gyrus responded stronger to pictures of food items with high than low energy density (i.e., it did show a main effect of energy density) but it did not respond stronger (or weaker) to large than small food portions (i.e. it did not show a main effect of portion size; English et al., 2017). Finally, also (very) large structural neuroimaging studies (Janowitz et al., 2015; Veit et al., 2014; Yokum et al., 2012) found correlations between morphological parameters of fusiform gyrus and temporal cortex (e.g. for GM volume or cortical thickness) on one hand and markers of obesity such as heightened BMI on the other. Consequently, we do not believe that the stimulus set used in the cue-reactivity paradigm hampered the interpretability of findings made Analysis 1.

A final point that should be discussed is the fact that the T-3 (T0) fMRI measurements took place shortly after diet onset (briefly before diet offset) in few participants. These deviations were caused by organizational reasons such as scanner or physician availability. For two reasons, we do not believe that these deviations severely hamper our interpretation of results. First, the strong overlap of coordinates identified in supplementary fMRI analysis S2 and coordinates identified in Analysis 1 and 3 in the main text makes this assumption unlikely. For example, 98% of all coordinates showing a stable neural response to food compared to control pictures in the period from T0 to T36 in supplementary fMRI Analysis S2 were also found in Analysis 1 for the full study period. In the supplementary fMRI analysis, however, only five out of 78 fMRI data sets (cf. Fig. 1) violated the experimental protocol since they were acquired before the end of the twelve weeks diet. Moreover, these five scans were all assessed during the final four of the twelve weeks of the diet. In this final period, the diet was characterized by a comparably normal caloric supply (1500 kilocalories per day) and comprised only few treatment components (provision of cooking advices, instructions for behaviour modifications, and weekly group meetings). Thus, this final period of the diet did not differ strongly from the period immediately after the end of the diet. Second, and more important, the focus of this study is not on neural substrates of decision-making or predictability of dietary variables in small and specific time windows. Instead, we were mainly interested in neural processes covering years and their impact on long-term dietary success due to the high clinical importance of such protracted processes. Thus, even when an fMRI scan was acquired slightly too late or too early compared to the time points predefined by the experimental protocol, it was still a valid indicator of these long-term processes since all measurements were made in a clinically important time period. This notion is further supported by the fact that we used piecewise linear models and LME regression for the analysis of longitudinal effects as this combination of techniques is very well suited to model such timing inaccuracies (Bernal-Rusiel et al., 2013a; 2013b; Wagner et al., 2002). Consequently, for these reasons, we do not believe that the timing deviations from the exact measurement protocol hamper the interpretability of results.

Taken together, we evaluated the impact of the interactions between Pavlovian and goal-directed decision-making systems on dietary treatment success based on the largest longitudinal fMRI data set currently evaluated in obesity research. The results suggest that a suitable interplay between these systems is a prerequisite for successful long-term maintenance of weight loss achieved in a short-term diet.

Conflicts of interest

The authors declare no financial interest or any conflict of interest in this work.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.neuroimage.2018.09.058.

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